

Yiyang Du

Carnegie Mellon University, Pittsburgh, PA, US
yiyangd@andrew.cmu.edu $\diamond$ GitHub : ADu2021 $\diamond$ adu2021.github.io

EDUCATION

Carnegie Mellon University, Pittsburgh, PA, US 2025.8 - current

Ph.D student, Language and Technologies Institute, School of Computer Science

Advisor: **Prof. Chenyan Xiong**

Direction: **Embodied Foundation Models, Multimodal Large Language Models**

Tsinghua University, Beijing, China 2021.9 - 2025.6

B.Eng in Computer Science and Technology (**Graduate with Honors**)

GPA: 3.98/4 (**Rank 1/173**)

Honors and Scholarships:

- Outstanding Graduate
- CCF Elite Collegiate Award
- Champion of China International College Students' Internet+ Innovation and Entrepreneurship Competition, Beijing Division
- First Prize, CSP-S (Equivalent to the original National Olympics in Informatics in Provinces, NOIP)

Achieved **A+** grades in the following courses:

- Linear Algebra
- Discrete Mathematics
- Algorithm Design and Complexity Analysis
- Programing and Training
- Introduction to Computer Systems
- Cybersecurity Fundamentals

PUBLICATIONS AND RECENT MANUSCRIPTS

Yiyang Du, Zhanqiu Guo, Xin Ye, Liu Ren, and Chenyan Xiong. **EmbodiedMidtrain: Bridging the Gap between Vision-Language Models and Vision-Language-Action Models via Mid-training.** *Under Review at COLM 2026. FMEA Workshop @ CVPR 2026 **Best Paper Award**.*

Yiyang Du, Xiaochen Wang, Chi Chen, Jiabo Ye, Yiru Wang, Peng Li, Ming Yan, Ji Zhang, Fei Huang, Zhifang Sui, Maosong Sun, and Yang Liu. **AdaMMS: Model Merging for Heterogeneous Multimodal Large Language Models with Unsupervised Coefficient Optimization.** *In Proceedings of Computer Vision and Pattern Recognition Conference (CVPR) 2025.*

Chi Chen*, **Yiyang Du***, Zheng Fang, Ziyue Wang, Fuwen Luo, Peng Li, Ming Yan, Ji Zhang, Fei Huang, Maosong Sun, and Yang Liu. **Model Composition for Multimodal Large Language Models.** *In Proceedings of Annual Meeting of the Association for Computational Linguistics (ACL) 2024.*

RESEARCH EXPERIENCES

Embodied Foundation Models 2025.8-current

Advised by Prof. Chenyan Xiong

Language Technologies Institute, Carnegie Mellon University

- Bridge the gap between the Vision-Language Model and Vision-Language-Action Models via mid-training, identify and analysis the data distribution gap between VLM data and VLA data to provide a better initialization for embodied training.
- Develop full pipeline of training system for Vision-Language-Action models, from dataset curation to multi-stage training, for robotics and autonomous driving.
- Investigate World Models and World Action Models for action-conditioned future prediction.

Model Merging for Multimodal LLMs 2023.10-2025.3

Advised by Prof. Yang Liu

AIR, Tsinghua University

- Proposed a model composition framework to fuse the modalities of multimodal large language models (MLLMs) without the burden of additional training.
- Designed parameter decoupling strategy and adaptive parameter adjustment algorithm to improve model merging performance in MLLMs by mitigating the parameter interference problem.
- Demonstrated the effectiveness of the framework by merging text, audio, vision, video, and 3D point cloud modalities into one model, while preserving the capabilities of each modality.

- Advanced the model merging strategy from identical architectures to heterogeneous architectures.

Large-Scale Pretraining of LLMs

2023.2 - 2023.7

Supervised by Prof. Maosong Sun

DeepLang AI, Beijing, China

- Investigated the influence of pretraining data on model performance to determine the optimal corpus composition regarding source, quality, and knowledge distribution.
- Studied the deployment of large-scale training on multi-node clusters with Megatron-LM training framework.

Semantic Retrieval

2022.11 - 2023.3

Supervised by Prof. Maosong Sun

THUNLP Lab, Tsinghua University

- Applied language model-based semantic retrieval strategy on English and Chinese dictionary corpus.
- Proposed WantQuotes, an online reverse dictionary system that helps users retrieve sentences or phrases based on their meaning.

ACHIEVEMENTS

Outstanding Graduate (2%), <i>Tsinghua University</i>	2025.6
Elite Collegiate Award (100 students nationwide), <i>China Computer Federation</i>	2024.10
National Scholarship (2%), <i>Ministry of Education</i>	2023.11
Comprehensive Outstanding Scholarship (Highest Award), <i>Tsinghua University</i>	2023.11
Champion of Beijing Division, <i>China International College Students' Internet+ Innovation and Entrepreneurship Competition</i>	2023.8
National Scholarship (2%), <i>Ministry of Education</i>	2022.11
Comprehensive Outstanding Scholarship (Highest Award), <i>Tsinghua University</i>	2022.11
First Prize, 2019 CSP-S Non-Professional Software Capability Certification (Equivalent to the original National Olympics in Informatics in Provinces, NOIP), <i>China Computer Federation</i>	2019.11

SERVICES

Teaching Assistant, Qinghai University	2022 Fall
· Provided teaching support and prepared assignments for the course <i>The Foundation of Programming (Advanced Level)</i> .	
Volunteer for Beijing 2022 Winter Olympics	2022.2
· Provided service to the audience at Wukesong Cadillac Arena.	
Volunteer Reviewer	
· ACL 2025	

PROFESSIONAL SKILLS

Skilled in using C++ and Python for research tasks

Deep learning libraries & frameworks: Megatron-LM, DeepSpeed, etc.

High performance computing with CUDA

TOEFL iBT: 105 out of 120 (Speaking 23)